Final Project - Part B: LLM-Tournament Writeup

208288647, 318170917, 026548446

1. Model Description and Experimental Setting

a. Chosen Model: LLM-Tournament Re-ranker (Run 2, final score on test: 0.4044).

b. Motivation: We developed a tournament-based LLM approach to leverage the zero-shot reasoning capabilities of generative models, moving beyond simple keyword matching to comparative relevance judgment. Inspired by HM3, we went in an LLM-centric direction, capitalizing on their generative reasoning to handle complex retrieval tasks - like query expansion, mock-doc generation, and listwise re-ranking.

c. Methodology and Techniques (the multi-stage process):

- **Candidate Generation:** An initial pool of 1,000 documents is retrieved using a weighted fusion of BM25 and QL with RM3 expansion. This mimics the pooling done to annotate the collection, ensuring high recall at a reasonable depth (in the training we used top 400 docs and in the test top 300).
- **Passage Extraction:** To handle LLM token limits and document noise, we used a sliding window (150 words, 75-word stride). A Cross-Encoder (ms-marco-MiniLM-L-6-v2) identifies the most relevant "best window" to represent the document. We found using the original query (not expanded one) was best here.
- **Seeded Tournament Logic:** We implemented a tournament where GPT-4o-mini acts as a judge. Documents were compared in batches of 4 (and not pairwise due to usage limits), sent with the original query and a prompt - *"You are a relevance judge ... return one document id, e.g. 2".* Winners earn exponential points based on the round $score = \sum_{r=1}^{n} 2^{(r-1)}$ thus "winners" are marked clearly. The tournament's 1st round batches were "Seeded", then random shuffling for the rest of the rounds.
- **Fusion:** Finally, we combined the normalized LLM and base scores $(W_{LLM} = 0.85, W_{Base} = 0.15)$.

d. Experimental Setting: Experiments were conducted using Pyserini on the ROBUST04 dataset, with 50 queries used for grid-search optimization.

- **Refined Expansion:** Attempts to expand queries during Passage Extraction introduced significant noise. Shifting expansion to the Base-Retrieval phase improved recall without degrading extraction precision.
- **Tournament Design:** Re-ranking was restricted to the top 300 documents. We compared three strategies: Seeded (e.g., 1st vs. 300th,299th,298th), Leapfrog (e.g., 1st vs. 100th,200th,300th), and Random.
- **Optimal Configuration:** The "Seeded" tournament with batch size 4 setup performed best. By seeding the first round, we protected high-precision documents (we got from our original base approach) from early elimination. Subsequent rounds used random shuffling to encourage semantic diversity.
- **Final Scoring:** We fused the base retrieval scores with LLM tournament rankings (normalized by the maximum score) for the top 300 documents; the remaining documents kept their base order. Due to this normalization method, the final weights favored the tournament scores.

2. Related Work - Academic Summaries:

1. Qin, Z., Jagerman, R., Hui, K., Zhuang, H., Wu, J., & Bendersky, M. (2023). Large language Models are Effective Text Rankers with Pairwise Ranking Prompting. *arXiv*. https://arxiv.org/abs/2306.17563
2. Sun, W., Yan, L., Ma, X., Wang, S., ... & Ren, Z. (2023). Is ChatGPT Good at Search? Investigation of Large Language Models as Re-Ranking Agents. *arXiv*.https://arxiv.org/abs/2304.09542
3. Nogueira, R., & Cho, K. (2019). Passage re-ranking with BERT. *arXiv*. https://arxiv.org/abs/1901.04085

Related Work & Comparison: Qin et al. introduced the concept of ranking as a pairwise input generation task using LLMs to reduce position bias. Like them, we treat rankings as a comparative "choice" in a multi-stage tournament. **Sun et al.** demonstrated the superior performance of listwise ranking but noted the difficulty of handling long contexts within LLM token limits. Similarly, we addressed these trade-offs; while pairwise approaches were initially safer, we successfully adapted a listwise strategy with a batch size of 4 to balance ranking quality with API constraints. Finally, **Nogueira and Cho** established the standard of using BERT-based Cross-Encoders for high-precision re-ranking. Unlike their approach of using the Cross-Encoder for the final score, we utilized it only for optimal passage selection, reserving the generative LLM for the final, complex comparative reasoning.